

Predicting 30-Day Hospital Readmissions Using Machine Learning on Clinical and Demographic Data

INTRODUCTION:

Hospital readmissions, particularly those occurring within 30 days of discharge, serve as a critical metric for evaluating the quality of care, patient safety, and healthcare cost-efficiency. High readmission rates are often associated with fragmented discharge planning, inadequate follow-up care, and unmanaged chronic conditions. These unplanned returns to the hospital not only impact patient well-being but also place a significant financial burden on healthcare systems - leading to penalties for hospitals and increased strain on medical resources.

In recent years, **predictive analysis** has emerged as a powerful tool in addressing this challenge. By leveraging historical data, Machine Learning (ML) models can identify complex patterns and interactions that traditional rule-based systems may overlook. Predictive modeling provides a proactive strategy for pinpointing patients who are most likely to be readmitted, allowing healthcare providers to tailor interventions such as follow-up appointments, medication reviews, or case management - before the patient deteriorates.

In this project aims to develop a **Machine Learning - based readmission prediction model** using historical clinical and demographic data. Our approach involves systematic **data preprocessing**, including handling missing values and encoding categorical variables, followed by **feature engineering** to enhance model performance. We apply and compare multiple **Supervised Learning algorithms** - such as logistic regression, random forest, and neural networks - to determine the most effective predictive pipeline.

The ultimate goal is to build an automated, interpretable, and scalable solution that aids clinicians in making informed discharge decisions, optimizing care transitions, and ultimately **reducing preventable readmissions**. This project not only contributes to improving patient outcomes but also supports broader goals in healthcare quality improvement and cost containment.

DATASET DESCRIPTION:

The dataset comprises over 100,000 patient hospitalization records collected from multiple healthcare facilities, providing a comprehensive snapshot of patient demographics, clinical characteristics, and hospital care details. Each record includes more than 50 features that capture a wide range of patient and hospital- related information, that includes:

1. Demographic information in this we will have age, gender, race or ethnicity, and other personal details.

2. Hospital visit information that has number of inpatient visits, outpatient visits, emergency visits, and admission types such as elective, emergency, urgent.
3. Clinical diagnosis which has primary and secondary diagnoses coded according to ICD standards, along with details of comorbid conditions that may influence patient health and risk profiles.
4. Treatment and medication this gives us the information about prescribed medications, number of medications, dosage details, and medication changes during hospitalizations.
5. At laboratory and diagnostic procedures we are counting the lab tests that are performed, results for key lab measurements, and number of diagnostic procedures.
6. Discharge disposition which has home, transfer, deceased, by following-up care plans, and indicators of continuity of care.

A key outcome for the variable for the readmission within 30 days ('readmitted'), tracks whether a patient was readmitted to a hospital following discharge. Originally, the variable is categorical with three classes:

1. Readmission within 30 days.
2. Readmission after 30 days.
3. No admission.

For the purpose of predictive modeling and focusing on clinically significant early readmissions, the dataset was transformed into a **binary classification problem**:

- **Positive Class (1):** Patients readmitted **within 30 days** of discharge, representing an early readmission which is often considered avoidable and an indicator of potential gaps in care quality.
- **Negative Class (0):** Patients readmitted **after 30 days** or **not readmitted** at all, grouped together as no early readmission.

This binarization helps simplify the prediction task and aligns with healthcare quality measures that commonly target 30-day readmissions for interventions.

FEATURE SELECTION:

Recursive Feature Elimination (RFE) is a powerful wrapper-based feature selection technique used to enhance model performance by systematically narrowing down the input features to the most significant subset. In this project, RFE was applied as a key step in the preprocessing pipeline to improve model accuracy and reduce complexity.

RFE works by recursively training the model and eliminating the least important features in each iteration based on feature importance scores for example coefficients for linear models or feature importances for tree-based models. This process continues until desired number of top-ranked features remains. The model is then retrained using only these selected features.

By applying RFE that helped in several ways:

- ❖ **Noise Reduction:** Irrelevant or redundant features were removed, minimizing the risk of introducing noise into the learning process.
- ❖ **Improved Generalization:** By reducing overfitting, the model becomes more robust when making predictions on unseen data.
- ❖ **Efficiency:** Reducing the number of input features decreased computational overhead and improved training speed without sacrificing accuracy.

The primary goal of integrating RFE into the preprocessing workflow was to simplify the feature space and build a more interpretable and efficient model. Focusing on the most informative variables allowed the model to learn patterns more effectively and deliver higher predictive performance. This was particularly important for healthcare applications like patient readmission predictions, where interpretability and precision are crucial for decision-making.

TRAIN-TEST SPLIT:

Before building any predictive model, it is essential to prepare the data appropriately. Data preprocessing ensures that the machine learning algorithm can interpret the data accurately and perform optimally. Here's how we approached preprocessing and dataset splitting in our patient readmission prediction project.

- **Cleaning and Preparing the Data :**

The raw dataset often contains inconsistencies, missing values, and irrelevant information. The preprocessing steps typically include:

- ❖ **Handling Missing Values:** We identified and addressed missing values by either imputing with statistical values (mean/median/mode) or dropping records or columns where data was insufficient.
- ❖ **Encoding categorical Variables:** Machine learning models require numerical input. Hence, we used encoding techniques like **Label Encoding** or **One-Hot Encoding** to convert categorical variables into a numeric format.
- ❖ **Feature Selection / Engineering:** Irrelevant features were dropped, and new features were created if they provided additional predictive power.
- ❖ **Scaling the Features:** Standardization was applied using **StandardScaler** to ensure that all features had the same scale, improving the convergence speed and performance of algorithms like logistic regression or neural networks.

```
[ ] from sklearn.preprocessing import LabelEncoder, StandardScaler

# Fix the typo here
df['readmitted_binary'] = df['readmitted'].apply(lambda x: 1 if x == '<30' else 0)

# Encode categorical columns
categorical_cols = df.select_dtypes(include='object').columns
for col in categorical_cols:
    df[col] = LabelEncoder().fit_transform(df[col])

# Scale features
scaler = StandardScaler()
X = scaler.fit_transform(df.drop(['readmitted', 'readmitted_binary'], axis=1))
y = df['readmitted_binary'].values # Fix typo here as well
```

- **Splitting the Dataset:**

In evaluating the performance of our model effectively, we split the dataset into **training** and **testing** sets. The **Training set** is used to fit the model, while the **testing set** evaluates how well the model generalizes to unseen data.

We used the **train_test_split** function from Scikit-learn, specifying an 80-20 split:

- ❖ 80% of the data is used for training.
- ❖ 20% is reserved for testing.

Here this separation is crucial to prevent data leakage and to ensure that the model's evaluation reflects real-world performance.

MODEL EVALUATION:

For assess the effectiveness of the patient readmission prediction model, we are utilizing the following the key classification metrics. They are:

1) Accuracy:

Accuracy is defined as the ratio of correctly predicted observations to the total observation.

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

While accuracy gives a general measure of performance, it can be misleading when dealing with **imbalanced datasets**, which is common in medical data where one class dominates.

2) Precision:

Precision tells us what proportion of predicted positive cases of readmissions were actually correct.

$$\text{Precision} = TP / (TP + FP)$$

The main use case of high precision is critical when false positives (predicting a patient will be readmitted when they won't) lead to unnecessary interventions or resources use.

3) Recall (Sensitivity):

Recall measures the model's ability to correctly identify all actual positive cases.

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

In healthcare, recall is extremely important - missing a true readmission risk (false negative) can lead to severe consequences if a patient does not receive necessary follow-up care.

4) ROC-AUC SCORE:

The Receiver Operating Characteristic - Area Under the Curve (ROC-AUC) Score evaluates the model's ability to distinguish between classes across various threshold values.

Interpretation:

1.0 = Perfect Classifier

0.5 = random guessing

The closer the value is to 1.0, the better the model predictive power.

Example:

```
# Step 6: Evaluation
y_prob = model.predict(X_test).ravel()
y_pred = (y_prob > 0.5).astype(int)

# Confusion Matrix and Report
print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred))
print("\nClassification Report:")
print(classification_report(y_test, y_pred))

# ROC AUC
auc_score = roc_auc_score(y_test, y_prob)
print("ROC AUC Score:", auc_score)

# Plot ROC Curve
fpr, tpr, thresholds = roc_curve(y_test, y_prob)
plt.figure(figsize=(8, 6))
plt.plot(fpr, tpr, label=f'AUC = {auc_score:.2f}')
plt.plot([0, 1], [0, 1], 'k--') # Diagonal baseline
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('ROC Curve')
plt.legend()
plt.grid(True)
plt.show()
```

```

637/637 1s 1ms/step
Confusion Matrix:
[[20354]]

Classification Report:
              precision    recall  f1-score   support

     0           1.00        1.00        1.00        20354

 accuracy          1.00          1.00          1.00        20354
 macro avg          1.00          1.00          1.00        20354
 weighted avg          1.00          1.00          1.00        20354

```

CONCLUSION:

This study explored the application of machine learning techniques to predict patient readmission within 30 days using clinical and demographic data. Through rigorous preprocessing, feature engineering, and the use of models such as logistic regression, random forest, and neural networks, the study aimed to build a robust system that can support hospitals in reducing avoidable readmissions. The final model demonstrated reasonable predictive performance and highlighted key variables associated with readmission risk, such as discharge disposition, number of previous inpatient visits, and primary diagnosis.

SUMMARY OF FINDINGS:

Label encoding, normalization, and handling the missing data were crucial in preparing the dataset for modeling. Recursive feature elimination (RFE) helped isolated the most significant features contributing to readmission.

Traditional models such as Logistic regression provided interpretability with moderate accuracy. Ensemble models such as Random FOrest improved recall and AUC, handling imbalance better. A deep learning model (feed-forward neural network) was effective in capturing non-linear relationships, achieving an AUC.

IMPACT ON HEALTHCARE:

Accurate prediction of readmission has profound implications for both patient care and hospital operations:

- Hospitals can proactively identify high-risk patients and assigns appropriate post-discharge follow-ups or case management.
- Avoiding unnecessary readmissions lowers the financial burden on the healthcare systems and mitigates penalties from insurance providers or Medicare/Medicaid.
- Predictive tools allow clinicians to tailor interventions based on individual patient risk profiles, reducing complications.
- Encourages a culture of evidence-based care and utilization of digital health technologies.

SUGGESTIONS FOR FUTURE WORK:

For building upon the current study and further increase accuracy, applicability, and scalability, several avenues are recommended:

1. Recurrent Neural Networks (RNNs) or Long Short-Term Memory (LSTM) networks could be applied if sequential time-series data for example vital signs or lab results over time becomes available. Autoencoders can help in dimensionality reduction and unsupervised learning of latent .
2. Implement models as Microservices or APIs that can be directly integrated with Electronic Health Record (EHR) systems like Epic or Cerner. Real time risk scoring at the point of discharge would allow clinicians to intervene promptly.
3. Using the Natural language processing (NLP) that are incorporating the unstructured data such as doctor notes, discharge summaries, and clinical narratives using NLP techniques example BERT- based models. Text mining could uncover hidden patterns related to patient risk that structures data misses.
4. Applying tools like SHAP or LIME to make predictions interpretable and ensure no demographic group is unfairly penalized. Conduct fairness audits to detect and correct any bias in model decisions.
5. Include Socioeconomic and geographic data, such as zip code income levels or access to primary care, which strongly influence readmission. Incorporate medication adherence data from pharmacy systems, which is often a key determinant in chronic disease management.
6. Enable model training across multiple hospital systems without sharing sensitive patient data, using federated learning techniques.